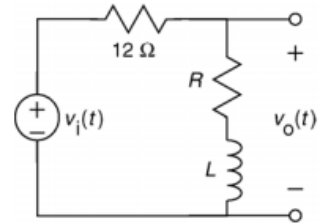
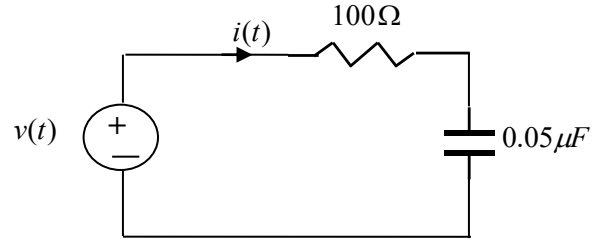
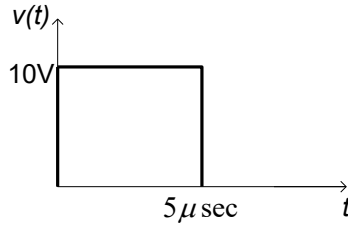


1. The step response $v_o(t)$ of the circuit is Determine R and L.

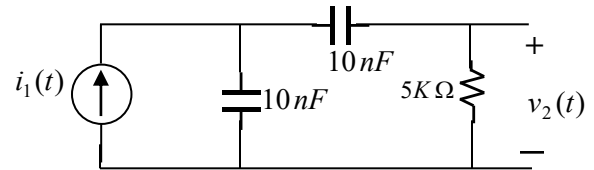
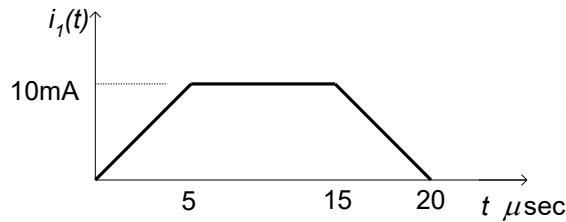
$$v_o(t) = 0.5(1 + e^{-4t})u(t) \text{ V}$$



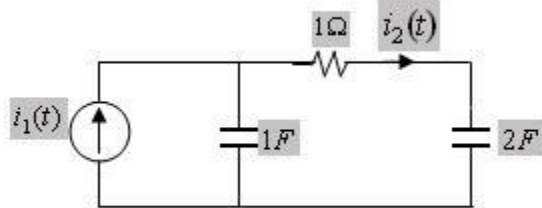
2. A voltage pulse shown in Figure is applied to the RC network shown. Find the current using network function approach.



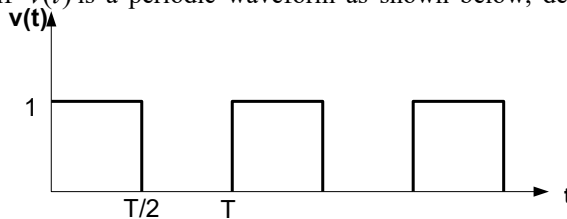
3. Find $Z_{12}(s)$ and determine $v_2(t)$. $i_1(t)$ is shown in Figure.



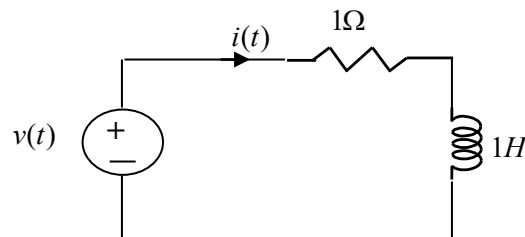
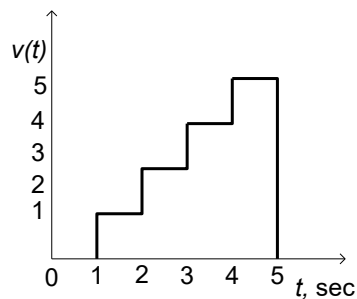
4. Find $\alpha_{12}(s)$ and determine $i_2(t)$. The input $i_1(t) = 2e^{-3t}$



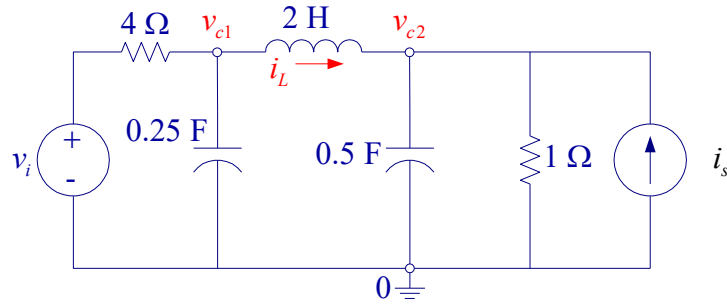
5. In the Figure of problem no.2 if $v(t)$ is a periodic waveform as shown below, determine $i(t)$. Assume $R = 1\Omega$ and $C = 1F$.



6. Using the network function approach, find the response $i(t)$. The input voltage $v(t)$ is shown in Figure.



7. Write the state equation for the network shown in Figure A.1.

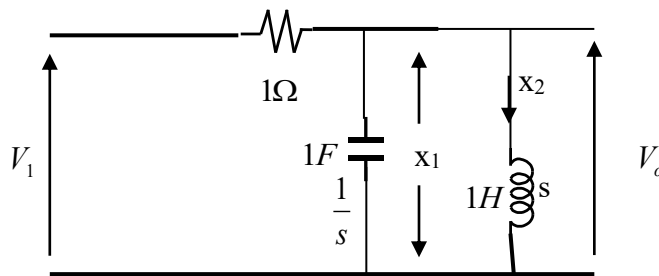


8. Derive the state-space model for each of the following differential equations.

1. $\ddot{y} - 2\ddot{y} + 3\dot{y} + 2y = u$

2. $\ddot{y} + 4\dot{y} - y = 3\dot{u} - 2u$

9. Obtain the state variable representation of the following circuit and solve the state equations to obtain unit step response of the circuit. Find the eigenvalues of A matrix.



10. A system is described by the following state-space equations

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = [8 \quad 1] \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}. \text{ Obtain the system transfer function.}$$

11. Find all equilibrium points and determine the nature of equilibrium points.

(a) $\dot{x}_1 = x_2, \quad \dot{x}_2 = -x_1 + \frac{x_1^3}{6} - x_2$

(b) $\dot{x}_1 = -x_1 + x_2, \quad \dot{x}_2 = -0.1x_1 - 2x_2 - x_1^2 - 0.1x_1^3$

12. Determine the steady-state output of the network shown in question 9, when the circuit is subjected to the following sinusoidal inputs:

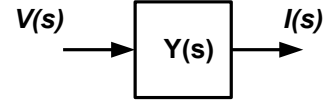
i) $10 \sin 2t$

ii) $10 \sin 2t + 20 \sin 5t$

Solution of question no. 6,

Answer: In the network input is $v(t)$ and output is $i(t)$. So the network function is $Y(s)$.

$$I(s) = Y(s)V(s).$$



$v(t)$ is shown in the Figure. We can write $v(t)$ as

$$v(t) = u(t-1) + u(t-2) + u(t-3) + u(t-4) - 4u(t-5)$$

$$V(s) = (e^{-s} + e^{-2s} + e^{-3s} + e^{-4s} - 4e^{-5s}) \frac{1}{s}$$

$$Z(s) = s + 1, \text{ So } Y(s) = \frac{1}{s+1}, I(s) = \frac{1}{s+1} (e^{-s} + e^{-2s} + e^{-3s} + e^{-4s} - 4e^{-5s}) \frac{1}{s}$$

$$L^{-1}\left[\frac{1}{s(s+1)}\right] = 1 - e^{-t},$$

$$i(t) = [1 - e^{-(t-1)}]u(t-1) + [1 - e^{-(t-2)}]u(t-2) + [1 - e^{-(t-3)}]u(t-3) + [1 - e^{-(t-4)}]u(t-4) - 4[1 - e^{-(t-5)}]u(t-5)$$